

AIRLINE RESERVATION SYSTEM – FINAL COMPLETE FLOW

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SYSTEM OVERVIEW

Aspect	Description
Project Name	Airline Reservation System
Type	Mini Project / Console-based Application
Users	Passenger (User) + Administrator
Key Feature	Real Flight Seat Layout (3-3 Configuration)
Data Storage	Database (SQLite/MySQL) or JSON Files

● USER FLOW – COMPLETE (13 Steps)

🔒 Step 1: Authentication

Options available to user:

Option	Description
Login	Existing user with email & password
Sign Up	New user registration
Guest Booking	Book without account (limited features)

Sign Up Fields:

- Full Name
- Email ID (unique)
- Mobile Number
- Password
- Confirm Password

Login Fields:

- Email ID / Mobile Number
- Password

🔍 Step 2: Search Flights

Input Fields:

Field	Type	Example
From (Source)	Dropdown/Text	Chennai (MAA)

Field	Type	Example
To (Destination)	Dropdown/Text	Delhi (DEL)
Travel Date	Date Picker	15/04/2026
Number of Passengers	Number	2

Validation Rules:

- Date cannot be in past
- Source and destination cannot be same
- Passengers cannot be zero or negative

● Step 3: Flight List Display

Display Format:

text

```
=====
=====
| Flight No | Airline | Departure | Arrival | Duration | Price | Status |
=====
=====
| AI-202   | Air India | 08:30 AM | 11:45 AM | 3h 15m | ₹4,500 | Select |
| 6E-345   | IndiGo    | 10:00 AM | 01:15 PM | 3h 15m | ₹3,800 | Select |
| SG-101   | SpiceJet  | 02:00 PM | 05:30 PM | 3h 30m | ₹4,200 | Select |
=====
=====
```

Step 4: Fare Selection

Comparison Table:

Feature	Economy	Business
Cabin Baggage	7 kg	10 kg
Check-in Baggage	15 kg	25 kg
Meal	Not Included	Included
Seat Selection	At Check-in	Free Selection
Price	Base Price	Base Price × 1.8

User selects one → Base price determined

Step 5: Passenger Details

Loop for each passenger:

Field	Type	Validation
Full Name	Text	Only letters, min 3 chars
Age	Number	0-120 years
Gender	Dropdown	Male / Female / Other

Age Categories:

Category	Age Range	Fare Type
Infant	0-2 years	10% of adult fare

Category	Age Range	Fare Type
Child	2-12 years	75% of adult fare
Adult	12+ years	Full fare

Step 6: Contact Details

Field	Type	Validation
Phone Number	Number	10 digits, Indian format
Email ID	Text	Must contain @ and .com/.in

Confirmation Message:

"Ticket confirmation will be sent to your email and SMS"

Step 7: ★ SEAT SELECTION – REAL FLIGHT LAYOUT

3-3 Aircraft Configuration

Visual Layout:

text

	LEFT SIDE			RIGHT SIDE		
	A	B	C	D	E	F
Row 1	[]	[]	[]	[]	[]	[]
Row 2	[]	[]	[]	[]	[]	[]
Row 3	[]	[]	[]	[]	[]	[]
Row 4	[]	[]	[]	[]	[]	[]
Row 5	[]	[]	[]	[]	[]	[]

Seat Types & Icons

Seat Position	Type	Icon
A, F	Window	
C, D	Aisle	
B, E	Middle	—

Status Indicators

Status	Color	Symbol	Action
Available	Green		Can select
Booked	Red		Disabled, cannot select
Selected	Yellow		Current user's choice

Selection Rules

- One seat per passenger
- Cannot select already booked seats
- Seat map shown before selection
- User can change seat until payment

Step 8: Add-ons

Add-on	Cost	Options
Meal	+₹300	Veg / Non-Veg / No Meal
Extra Baggage (10kg)	+₹800	Yes / No

Display Format:

text

Add-ons Available:

1. Meal Preference:

- a) Veg (+₹300)
- b) Non-Veg (+₹300)
- c) No Meal (Free)

2. Extra Baggage:

- a) Yes (+₹800 for 10kg)
- b) No (Free)

 Step 9: Trip Summary

Complete Breakdown:

text

TRIP SUMMARY		
✈️ FLIGHT DETAILS		
Flight No:	AI-202	
Airline:	Air India	
Route:	Chennai (MAA) → Delhi (DEL)	
Date:	15 April 2026	
Time:	08:30 AM → 11:45 AM (3h 15m)	
👤 PASSENGER DETAILS		

SMS Template:

text

Airline: Booking confirmed!

PNR: PNR26A4B7C9

Flight: AI-202 MAA→DEL 15-Apr 08:30

Seat: A2

Amount: ₹6,100

Step 13: Booking Management

User Dashboard:

text

MY BOOKINGS

PNR: PNR26A4B7C9 |

Flight: AI-202 | Chennai → Delhi | 15-Apr-2026 |

Seat: A2 | Status: ✓ CONFIRMED |

[View] [Cancel] [Download Ticket] |

PNR: PNR26D9E2F3 |

|| | Flight: 6E-345 | Chennai → Bengaluru | 20-Apr-2026 | ||

|| | Seat: C1 | Status: ✓ CONFIRMED | ||

|| | [View] [Cancel] [Download Ticket] | ||

||

||

||

Cancel Functionality:

- Shows cancellation charges
- Refund calculation based on time of cancellation
- Confirmation before cancellation

● ADMIN FLOW – COMPLETE (9 Steps)

🔒 Step 1: Admin Login

Login Screen:

text

ADMIN LOGIN

Username:

Password:

Default Credentials:

Username

Password

admin

admin123

airline_admin

airline@2024

Step 2: Admin Dashboard

Dashboard Menu:

text

ADMIN DASHBOARD

Quick Stats:

- Total Flights: 12
- Total Bookings: 45
- Total Revenue: ₹2,45,000

- 1.  Add Flight
- 2.  Update Flight
- 3.  Delete Flight
- 4.  View All Flights
- 5.  View All Bookings

6.  Seat Management

7.  Logout

Step 3: Add Flight

Form Fields:

Field	Type	Example
Flight ID	Text	AI-202
Airline Name	Text	Air India
Source	Text	Chennai (MAA)
Destination	Text	Delhi (DEL)
Departure Date & Time	DateTime	15-04-2026 08:30 AM
Arrival Date & Time	DateTime	15-04-2026 11:45 AM
Economy Price	Number	4500
Business Price	Number	8100
Total Seats	Number	30

Step 4: Update Flight

Process:

1. Select flight from dropdown list
 2. Display current details
 3. Choose field to modify:
 - Departure/Arrival Time
 - Economy/Business Price
 - Available Seats
 4. Enter new value
 5. Confirm update → Save to database
-

Step 5: Delete Flight

Process:

1. Select flight from list
 2. Show warning: "This will delete all bookings for this flight"
 3. Confirmation: "Are you sure? (Y/N)"
 4. If Yes → Remove from database
 5. Success message: "Flight deleted successfully"
-

Step 6: View Flights

Display Format:

text

```
=====
=====
| ID   | Airline | Route   | Departure | Arrival | Eco | Biz |
=====
=====
```

| AI-202 | Air India | MAA → DEL | 08:30 AM | 11:45 AM | ₹4500 |
₹8100|

| 6E-345 | IndiGo | MAA → BLR | 10:00 AM | 11:15 AM | ₹2500 |
₹4500|

| SG-101 | SpiceJet | MAA → BOM | 02:00 PM | 04:30 PM |
₹3800 | ₹6800|

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Step 7: View Bookings

Display Format:

text

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| PNR | Passenger | Flight | Seat | Fare | Amount | Status |

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| PNR26A4B7C9 | Rajesh | AI-202 | A2 | Economy| ₹6,100 |
Confirmed |

| PNR26D9E2F3 | Priya | 6E-345 | C1 | Economy| ₹3,800 |
Confirmed |

| PNR26G5H8J1 | Amit | SG-101 | B3 | Business| ₹7,500 |
Confirmed |

=====

Step 8: Seat Management

Seat Map View:

text

Flight: AI-202 | Date: 15-Apr-2026

	A	B	C	D	E	F
Row 1						
Row 2						
Row 3						
Row 4						
Row 5						

Legend: ● Available | ● Booked

Options:

[1] Block Seat

[2] Unblock Seat

[3] Back to Dashboard

📁 Step 9: Logout

Logout Screen:

text

```
|| You have been logged out      ||
|| Redirecting to Home Page...   ||
|| Thank you!                    ||
```



DATABASE STORAGE DESIGN



DATABASE SCHEMA (MySQL / SQLite)

1. USERS Table

Column Name	Data Type	Constraints	Description
user_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique user ID
full_name	VARCHAR(100)	NOT NULL	User's full name
email	VARCHAR(100)	UNIQUE, NOT NULL	Email for login
phone	VARCHAR(15)	NOT NULL	Mobile number
password	VARCHAR(255)	NOT NULL	Hashed password
role	ENUM('user','admin')	DEFAULT 'user'	User role
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Registration date

2. FLIGHTS Table

Column Name	Data Type	Constraints	Description
flight_id	VARCHAR(10)	PRIMARY KEY	Unique flight code
airline	VARCHAR(50)	NOT NULL	Airline name
source	VARCHAR(50)	NOT NULL	Departure city
destination	VARCHAR(50)	NOT NULL	Arrival city
dept_datetime	DATETIME	NOT NULL	Departure date & time
arr_datetime	DATETIME	NOT NULL	Arrival date & time
eco_price	DECIMAL(10,2)	NOT NULL	Economy class price
biz_price	DECIMAL(10,2)	NOT NULL	Business class price
total_seats	INT	DEFAULT 30	Total seats in aircraft
available_seats	INT	NOT NULL	Seats available for booking

3. BOOKINGS Table

Column Name	Data Type	Constraints	Description
pnr	VARCHAR(12)	PRIMARY KEY	Unique PNR number
user_id	INT	FOREIGN KEY → USERS(user_id)	User who booked
flight_id	VARCHAR(10)	FOREIGN KEY → FLIGHTS(flight_id)	Booked flight
seat_number	VARCHAR(5)	NOT NULL	Selected seat
fare_type	ENUM('economy','business')	NOT NULL	Class selected
total_amount	DECIMAL(10,2)	NOT NULL	Total paid amount
booking_date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Booking timestamp
status	ENUM('confirmed','cancelled')	DEFAULT 'confirmed'	Booking status

4. PASSENGERS Table

Column Name	Data Type	Constraints	Description
passenger_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID
pnr	VARCHAR(12)	FOREIGN KEY → BOOKINGS(pnr)	Linked booking
name	VARCHAR(100)	NOT NULL	Passenger name
age	INT	NOT NULL	Passenger age
gender	ENUM('M','F','O')	NOT NULL	Gender

5. ADDONS Table

Column Name	Data Type	Constraints	Description
addon_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID
pnr	VARCHAR(12)	FOREIGN KEY → BOOKINGS(pnr)	Linked booking
meal_type	ENUM('veg','non-veg','none')	DEFAULT 'none'	Meal preference

Column Name	Data Type	Constraints	Description
extra_baggage	BOOLEAN	DEFAULT FALSE	Extra baggage (10kg)

6. SEATS Table (Seat Availability per Flight)

Column Name	Data Type	Constraints	Description
seat_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID
flight_id	VARCHAR(10)	FOREIGN KEY → FLIGHTS(flight_id)	Flight
seat_number	VARCHAR(5)	NOT NULL	Seat code (A1, B2, etc.)
is_booked	BOOLEAN	DEFAULT FALSE	Booking status
passenger_name	VARCHAR(100)	NULL	Booked passenger name